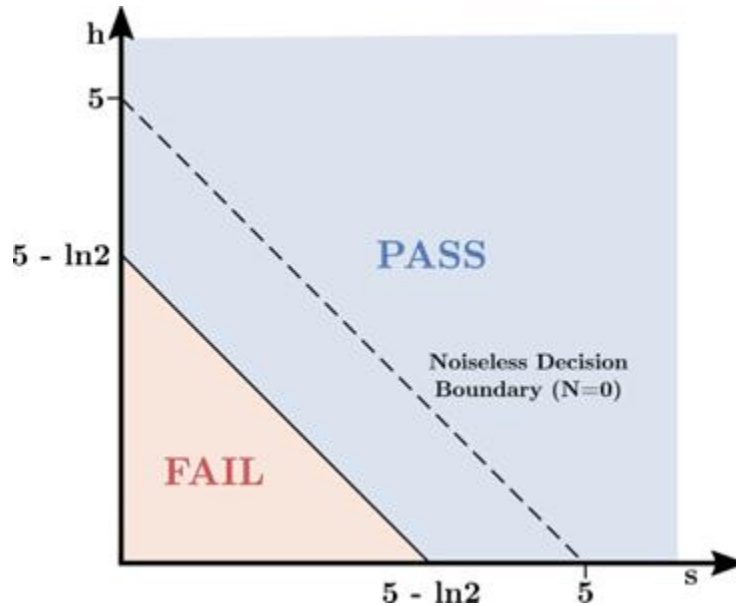


Question 1



Step 1: Computing the Bayes error.

$$\varepsilon^* = E[\min\{\eta(S, H), 1 - \eta(S, H)\}] = \iint_{\text{region1}} \eta(s, h) \cdot p(s, h) ds dh + \iint_{\text{region2}} (1 - \eta(s, h)) \cdot p(s, h) ds dh$$

Where Region 1: $s + h \leq 5 - \ln 2$ predicted as fail, and

Region 2: $5 - \ln 2 < s + h \leq 5$ predicted as pass.

Step 2: Evaluating Region 1 : ($s + h \leq 5 - \ln 2$):

$\eta(s, h) = e^{s+h-5}$ and $p(s, h) = e^{-(s+h)}$, because S and H are independent exponential distributions

$$\begin{aligned} \text{Integral 1} &= e^{s+h-5} \cdot e^{-(s+h)} \cdot \text{Area of region 1} \\ &= e^{-5} \cdot \text{Area of region 1} \\ &= e^{-5} \cdot \frac{(5-\ln 2)^2}{2} \end{aligned}$$

Step 3: Evaluating Region 2 ($5 - \ln 2 < s + h \leq 5$):

$$1 - \eta(s, h) = 1 - e^{s+h-5}$$

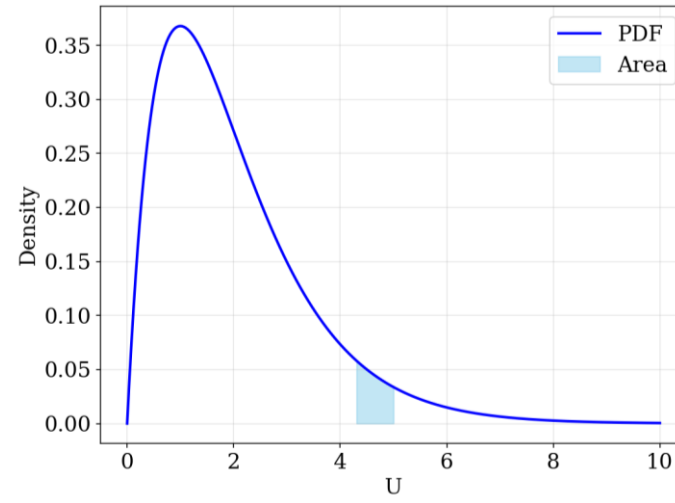
$$\begin{aligned}\text{Integrand 2} &= (1 - e^{s+h-5}) \cdot e^{-(s+h)} \\ &= e^{-(s+h)} - e^{-5}\end{aligned}$$

Now there are two terms: $\iint_{\text{region2}} e^{-(s+h)} dsdh + \iint_{\text{region2}} e^{-5} dsdh$

- Term 1:

$$\iint_{\text{region2}} e^{-(s+h)} dsdh = \iint_{\text{region2}} e^{-s} e^{-h} dsdh = \iint_{\text{region2}} p(s, h) dsdh:$$

The sum of n independent exponential random variables with parameter λ is a Gamma random variable with parameters λ and $t = n$.



Since $S \sim \exp(1)$, $H \sim \exp(1)$, $S + H \sim \Gamma(2,1)$:

$$P(a < S + H \leq b) = [1 - (1 + b)e^{-b}] - [1 - (1 + a)e^{-a}]$$

Substituting for a and b :

$$\begin{aligned}\text{Term 1} &= [1 - 6e^{-5}] - [1 - (6 - \ln 2)e^{-(5 - \ln 2)}] \\ &= -6e^{-5} + (6 - \ln 2) \cdot 2e^{-5}\end{aligned}$$

$$= e^{-5}(6 - 2 \ln 2)$$

- Term 2 $\iint_{region2} e^{-5} ds dh$:

$$\begin{aligned} \text{Term 2} &= e^{-5} * \text{Area of Region 2} \\ &= e^{-5} \left[\frac{5^2}{2} - \frac{(5 - \ln 2)^2}{2} \right], \text{ area of big triangle} - \text{small triangle.} \end{aligned}$$

Combining terms 1 and 2,

$$\text{Integral 2} = e^{-5}(6 - 2 \ln 2) + e^{-5} \frac{25 - (5 - \ln 2)^2}{2}$$

Step 4: Total error rate combining Regions 1 and 2 becomes:

$$\varepsilon^* \approx e^{-5} \cdot 10.633 \approx 0.0718 \approx 7.2\%$$